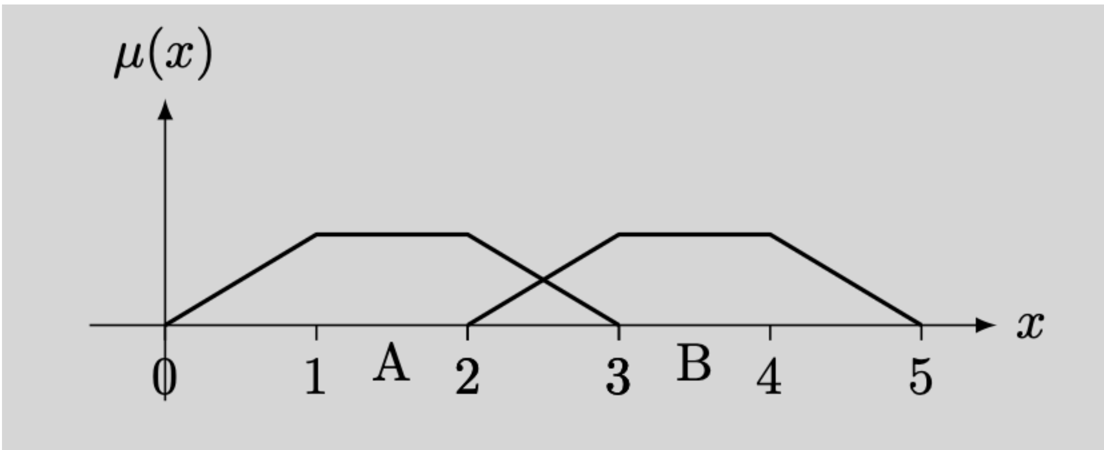
Fuzzy Systems and Evolutionary Computing 2022/23

Open Questions

- What is a linguistic variable? Give an example.
- What are fuzzification and defuzzification in a fuzzy rule based system? Which is the connection with a frame of cognition (aka, fuzzy partition)?
- Explain how fuzzy c-means works. If helpful, you can consider the two centroids c1 = (0, 1), c2 = (1,0), w = 2 and three objects x1 = (0.75, 0.5), x2 = (0, 0) and x3 = (0.3, 1.1).
- Define n-point crossover. Give an example.
- Mutation in EA: what is it? Which is its role? Give an example with integer representation
- Give a high-level (=no need of mathematical details) explanation of Particle Swarm Optimization

Exercises

1) Let us consider the two fuzzy sets A and B given in the following figure



   a) Give the equation of the two functions defining A and B
   b) Compute the intersection and union of f and g by means of a t-norm/t-conorm at your choice
   c) Compute the complement of A
   d) Compute support and core of G
2) Let us consider the fuzzy set *g* defined as g(5)=0.3, g(6)=0.5, g(7)=0.9, g(8)=1, g(9)=0.9, g(10)=0.5, g(11)=0.3 g(x) = 0 for all other values of x.
   a) Compute core and support of g
   b) Compute the complement of g
   c) Compute the cardinality of g
   d) Compute the α-cut of g with α=0.6
   e) Compute the entropy of g
   f) Compute the measure of fuzziness of g using the Euclidean distance
   g) Compute the ordered weighted average with $w_1= w_2 =w_3=2/14$, $w_4 = w_5= 1/14$, $w_6 = w_7=3/14$
   h) Compute the center of gravity of g

3) Given two propositions P and Q, with truth value v(P)=0.7 and v(Q)=0.4, compute
   a) The logical *and* using product t-norm
   b) The logical *or* using product t-conorm
   c) P→Q, using an implication at your choice
4) Let us consider a population of integer expressions (for instance (3+4)-(6+8)) with operations in {+,-,*,/} and numbers in {0,...,9}. Apply simple Genetic Programming with the following parameters
   a) Random initial population of 4 individuals generated with full method and depth = 3
   b) The fitness function is defined as $f(x) = x + 2$ with x the result of the expression represented by the tree
   c) Generate 4 new individuals: two are obtained by crossover of the two most fit parents and two by mutation of the other two parents
   d) Survivor selection: generational replacement
5) Let us consider a population of individuals represented by 6 binary digits. Apply simple Genetic Programming with the following parameters:
   a) Generate random initial population of 4 individuals
   b) Generate 4 new individuals: parents selected by roulette wheel selection, 2-point crossover and bit-wise mutation with $p_m = 1/6$
   c) Survivor selection: age based
   d) Create a new offspring with tournament selection and k=2

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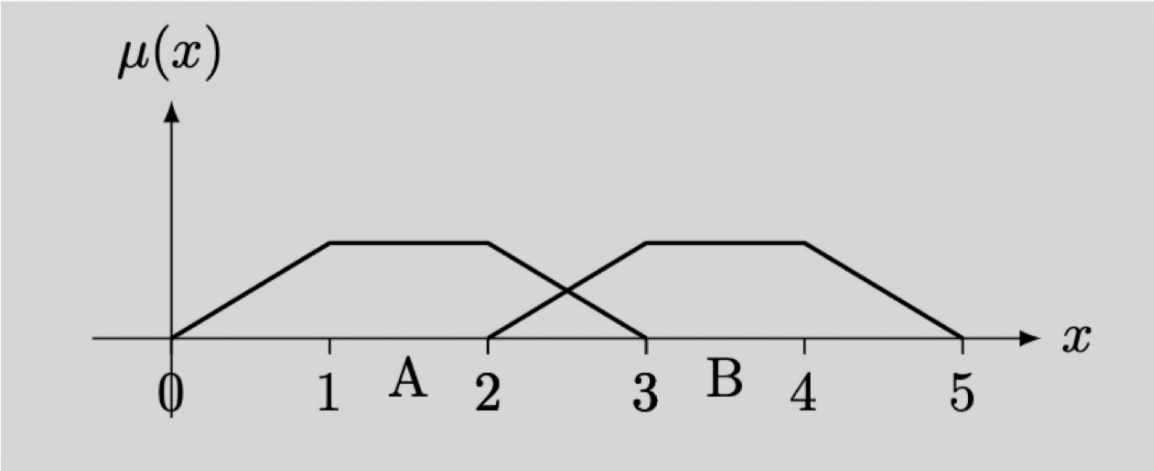

a) Give the equation of the two functions defining A and B
b) Compute the intersection and union of f and g by means of a t-norm/t-conorm at your choice
c) Compute the complement of A
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$$A = \begin{cases} x & \text{IF} \quad 0<x\leq 1 \\ 1 & \text{IF} \quad 1<x\leq 2 \\ 3-x & \text{IF} \quad 2<x\leq 3 \\ 0 & \text{OTHERWISE} \end{cases}$$

$$B = \begin{cases} x-2 & \text{IF} \quad 2<x\leq 3 \\ 1 & \text{IF} \quad 3<x\leq 4 \\ 5-x & \text{IF} \quad 4<x<5 \\ 0 & \text{OTHERWISE} \end{cases}$$

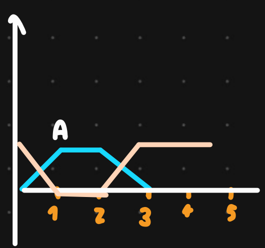
INTERSECTION (MIN)

$$(A\cap B)(x) = \begin{cases} x-2 & \text{IF} \quad 2<x\leq 2.5 \\ 3-x & \text{IF} \quad 2.5<x\leq 3 \\ 0 & \text{OTHERWISE} \end{cases}$$

UNION (MAX)

$$(A\cup B)(x) = \begin{cases} x & \text{IF} \quad 0<x\leq 1 \\ 1 & \text{IF} \quad 1<x\leq 2 \text{ OR } 3<x\leq 4 \\ 3-x & \text{IF} \quad 2<x\leq 2.5 \\ x-2 & \quad 2.5<x\leq 3 \\ 5-x & \quad 4<x\leq 5 \\ 0 & \text{OTHERWISE} \end{cases}$$

COMPLEMENT OF A



$$A^c(x) = \begin{cases} 1-x & \text{IF} \quad 0<x\leq 1 \\ 0 & \text{IF} \quad 1<x\leq 2 \\ x-2 & \text{IF} \quad 2<x\leq 3 \\ 1 & \text{OTHERWISE} \end{cases}$$

SUPPORT $B\ (>0) \to (2,5)$

CORE $B\ (=1) \to [3,4]$

2) Let us consider the fuzzy set *g* defined as g(5)=0.3, g(6)=0.5, g(7)=0.9, g(8)=1, g(9)=0.9, g(10)=0.5, g(11)=0.3 g(x) = 0 for all other values of x.
   a) Compute core and support of g
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- CORE (g) = {8}
- SUPPORT (g) = {5, 6, 7, 8, 9, 10, 11}

• g(5) = 0.3	• $g^c(5) = 0.7$
• g(6) = 0.5	• $g^c(6) = 0.5$
• g(7) = 0.9	$g^c(7) = 0.1$
• g(8) = 1	$g^c(8) = 0$
• g(9) = 0.9	$g^c(9) = 0.1$
• g(10) = 0.5	$g^c(10) = 0.5$
• g(11) = 0.3	$g^c(11) = 0.7$
• g(x) = 0	$g^c(x) = 1$

- CARDINALITY OF g
  0.3 + 0.5 + 0.9 + 1 + 0.9 + 0.5 + 0.3

- α CUT WITH α ≥ 0.6
  {7, 8, 9}

- ENTROPY OF g

$$H = -\big(0.3\log(0.3) + 0.5\log(0.5) + 0.9\log(0.9) + 1\log(1) + 0.9\log(0.9) + 0.5\log(0.5) + 0.3\log(0.7) + 0.7\log(0.7) + 0.5\log(0.5) + 0.1\log(0.1) + 0.1\log(0.1) + 0.5\log(0.5) + 0.7\log(0.7) + 1\log(1)\big)$$

FUZZYNESS W/ EUCLIDEAN

$$1 - \frac{\sqrt{(0.3-0.7)^2 + (0.5-0.5)^2 + (0.9-0.1)^2 + (1-0)^2 + (0.9-0.1)^2 + (0.5-0.5)^2 + (0.3-0.7)^2}}{\text{CARD } g}$$

ORDERED WEIGHTED AVR.

$w_1 = w_2 = w_3 = \frac{2}{14}$
$w_4 = w_5 = 1/14$
$w_6 = w_7 = 3/14$

$$1 \cdot \frac{2}{14} + 0.9 \cdot \frac{2}{14} + 0.9 \cdot \frac{2}{14} + 0.5 \cdot \frac{1}{14} + 0.5 \cdot \frac{1}{14} + 0.3 \cdot \frac{3}{14} + 0.3 \cdot \frac{3}{14}$$

CENTER OF GRAVITY

$$\frac{5 \cdot 0.3 + 6 \cdot 0.5 + 7 \cdot 0.9 + 8 \cdot 1 + 9 \cdot 0.9 + 10 \cdot 0.5 + 11 \cdot 0.3}{\text{CARD } g}$$

3) Given two propositions P and Q, with truth value v(P)=0.7 and v(Q)=0.4, compute
   a) The logical *and* using product t-norm
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   c) P→Q, using an implication at your choice

$\left.\begin{array}{l} v(P)=0.7 \\ v(Q)=0.4 \end{array}\right]$

$\boxed{\text{AND}}$

- PRODUCT $= 0.4 \cdot 0.7$
- MIN$(0.4, 0.7) = 0.4$
- ŁUKASIEW MAX$(0, P+Q-1)$

$\boxed{\text{OR}}$

- PRODUCT $= P+Q-PQ = 0.7 + 0.4 - (0.7 \cdot 0.4)$
- MAX$(0.4, 0.7) = 0.7$
- ŁUKASIEW MIN$(P+Q, 1)$

$\boxed{P \to Q}$

- GODEL: $Q = 0.4$
- GOGUGEN $Q/P = 0.4/0.7$
- ŁUKASIEW: $1-P+Q$

4) Let us consider a population of integer expressions (for instance (3+4)-(6+8)) with operations in {+,-,*,/} and numbers in {0,…,9}. Apply simple Genetic Programming with the following parameters
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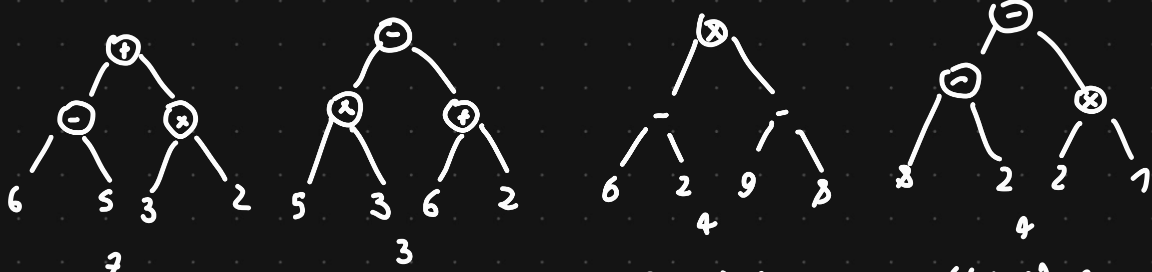
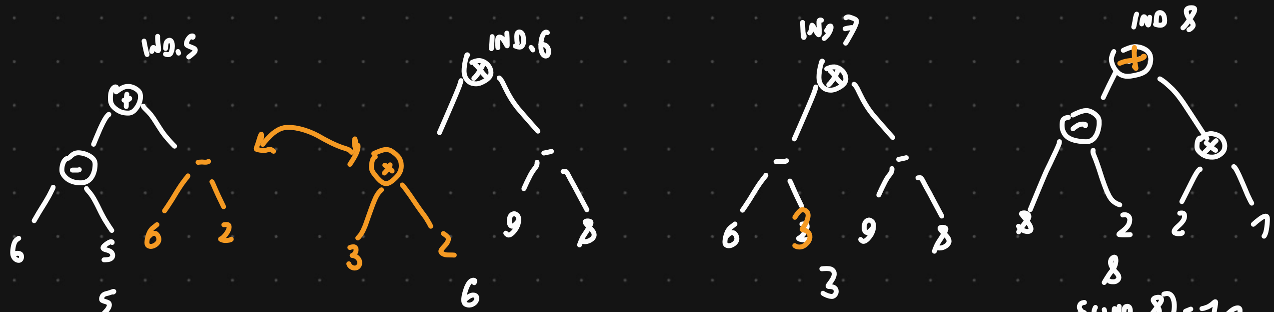


f(IND.1) = 9 f(IND.2) = 5 f(IND.3) = 6 f(IND.4) = 6

MOST FIT: IND 1 AND IND.3



f(IND.5) = 7 f(IND.6) = 8 f(IND.7) = 5 f(IND.8) = 10

GENERATIONAL REPLACEMENT

5) Let us consider a population of individuals represented by 6 binary digits. Apply simple Genetic Programming with the following parameters:

a) Generate random initial population of 4 individuals
b) Generate 4 new individuals: parents selected by roulette wheel selection, 2-point crossover and bit-wise mutation with $p_m = 1/6$
c) Survivor selection: age based
d) Create a new offspring with tournament selection and k=2

IND 1	010101	$2^0+2^2+2^4=21$	$P(IND.1) = 0.16$	CHOOSE 1 TIME
IND 2	101010	$2^1+2^3+2^5=42$	$P(IND\ 2) = 0.33$	CHOOSE 1 TIME
IND 3	000111	$2^0+2^1+2^2=7$	$P(IND\ 3) = 0.05$	CHOOSE 0 TIME
IND 4	111000	$2^3+2^4+2^5=56$	$P(IND\ 4) = 0.44$	CHOOSE 3 TIMES
		126		

IND.1	01|01|01	IND.5	011101	001101	$2^0+2^2+2^3 = 13$
IND.4	11|1|000	IND.6	110000	110100	$2^2+2^4+2^5 = 52$
IND.2	1|010|10	IND.7	111010	111010	$2^1+2^3+2^4+2^5 = 58$
IND.4	1|110|00	IND.8	101000	100010	$2^1+2^5 = 34$

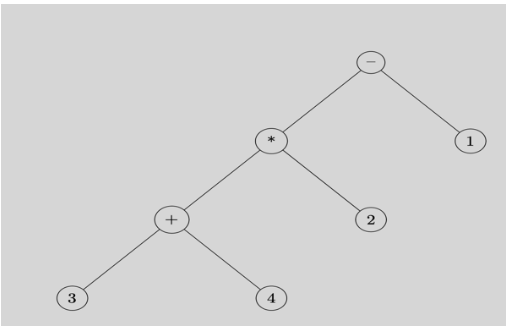
REMOVE IND 1, IND 2, IND 3 AND IND 4

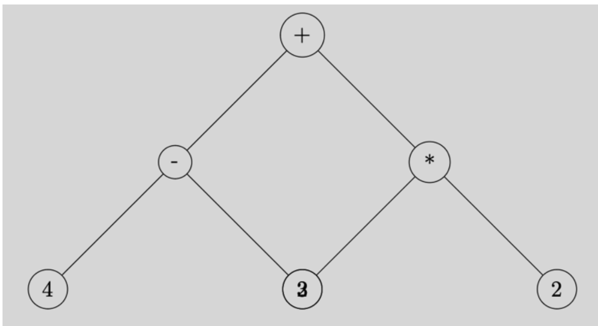
K=2 ⇒ KEEP ONLY IND 6 AND IND 7

Evolutionary computing - Exercises

- Given two individuals: 010101 and 101010
  - Apply *2-point* crossover
  - Apply $(\mu+\lambda)$ selection with f(x) = -x
- Given two individuals: 11111111 and 00000000
  - Apply *3-point* crossover
  - Apply $(\mu+\lambda)$ selection with f(x) = x
- Consider two permutations A,B,C,D,E,F and F,E,D,C,B,A
  - Apply *order 1* crossover on a subset of three elements
  - To one of the children apply *swap* mutation and to the other one *insert* mutation
- Repeat the previous exercise on the two permutations: 1,3,5,2,4,6 and 2,4,6,1,3,5.

- Consider the two trees below, representing, respectively, the two formulae ((3+4)*2)-1 and (4-3)+(2*2)





Apply crossover. Which one is the most fit individual if f(T) = |evaluate(T) -10|? [evaluate(T) is the result of evaluating the formula represented by the tree].

- Consider the two logical expressions $(x \wedge y) \vee (x \rightarrow y)$ and $(y \vee x) \rightarrow x$
  - Represent them by means of a binary tree
  - Compute their fitness with f(T)= number of truth assignments for which the expression represented by T evaluates to true
  - Apply crossover
- Let us consider an individual represented by 4 binary digit, for instance 1011, labeled from right to left as $x_o = 1, x_1 = 1, x_2 = 0, x_3 = 1$. The fitness function is defined as $f(x) = x_3 + 2x_2 + 6x_1 + x_0$. Apply simple GA evolution:
  - Generate random initial population of 4 individuals
  - Generate 4 new individuals: parents selected by roulette wheel selection, 1-point crossover and bit-wise mutation with $p_m = 1/4$
  - Survivor selection: age based
  - Create a new offspring with tournament selection and k=2

- Given two individuals: 010101 and 101010
  - Apply *2-point* crossover
  - Apply *(μ+λ)* selection with f(x) = -x

IND. 1   01|01|01 ⟶ 011001   IND.3

IND. 2   10|10|10 ⟶ 100110   IND.4

IND.1 $2^0+2^2+2^4 = 21$ f(IND.1) = -21

IND.2 $2^1+2^3+2^5 = 42$ f(IND.2) = -42

IND.3 $2^0+2^3+2^4 = 25$ f(IND.3) = -25

IND.4 $2^1+2^2+2^5 = 38$ f(IND.4) = -38

CHOOSE IND.1 AND IND.3

- Given two individuals: 11111111 and 00000000
  - Apply *3-point* crossover
  - Apply *(μ+λ)* selection with f(x) = x

IND.1   11|1|111|11 ⟶ IND.3   11011100

IND.2   00|0|000|00 ⟶ IND.4   00100011

IND.1 $2^0+2^1+2^2+2^3+2^4+2^5+2^6+2^7 = 297$

IND.2   0

IND.3 $2^2+2^3+2^4+2^6+2^7 = 220$

IND.4 $2^0+2^1+2^5 = 35$

CHOOSE IND.1 AND IND.3

- Consider two permutations A,B,C,D,E,F and F,E,D,C,B,A
  - Apply *order 1* crossover on a subset of three elements
  - To one of the children apply *swap* mutation and to the other one *insert* mutation

ABCDEF